

## WORK EXPERIENCE

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- **Amazon** August 2022 - June 2026
  - *Software Development Engineer II*
    - Designed and implemented a guardrail layer within a new LLM orchestration service connecting generative AI models with Amazon’s customer service chatbot. Validated customer inputs and model outputs to ensure safe and appropriate interactions, contributing to the launch of Amazon’s first LLM-powered customer support experience.
    - Designed and built reusable ML infrastructure using Amazon SageMaker to automate model training, testing, and deployment through a CI/CD pipeline. Deployed a model using this infrastructure and integrated it with the customer service chatbot to predict customer issues and route them to relevant support workflows, reducing average chatbot interaction time by 9% (440s to 400s) with no negative impact on other metrics.
    - Owned and implemented the integration of Amazon Demand-Side Platform (ADSP) into the Retailer Ad Service’s attribution service, extending existing Sponsored Products (SP) attribution to also support display and video advertising. Built unified data pipelines to ingest upstream SP and ADSP data, and developed attribution logic in Spark SQL within AWS Clean Rooms to jointly perform attribution across both programs.
    - Led scaling efforts for the Retailer Ad Service catalog and ingestion services to support Macy’s onboarding, handling catalogs with 2M+ offers uploaded as frequently as every two hours. Improved ingestion scalability and reliability, resolving service bottlenecks and enabling stable onboarding of the platform’s largest retailer to date.
- **Agency for Science, Technology and Research (A\*STAR)** Jun 2018 - Sept 2018
  - *Data Science Intern*
    - Developed **SeizNet**, a CNN-based seizure detection model, achieving 95.8% sensitivity with 0.17 false alarms/hour on EEG data from 29 patients; demonstrated that a 2-channel configuration achieved 93.3% sensitivity, outperforming a traditional SVM baseline using the full set of EEG channels.

## TECHNICAL SKILLS

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**Programming Languages:** Python, Java, TypeScript, Scala, JavaScript, SQL, C++

**Machine Learning and Data:** Large Language Models (LLMs), Spark, Keras, TensorFlow, Scikit-learn

**AWS Technologies:** CDK, CloudFormation, S3, Lambda, SQS/SNS, DynamoDB, SageMaker, Bedrock, Glue, Clean Rooms

**Others:** Linux, Docker

## EDUCATION

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- **University of Southern California** May 2022
  - *Master of Science in Computer Science, CGPA: 3.96/4.00*
    - Selected Courses: Machine Learning, Analysis of Algorithms, Database Systems, Applied Machine Learning for Games, Estimation Theory, Random Processes, Applied Natural Language Processing
- **Bilkent University** Jun 2019
  - *Bachelor of Science in Electrical and Electronics Engineering, CGPA: 3.93/4.00*

## HONORS & AWARDS

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- **Two-time Amazon org-wide hackathon winner among 20+ competing teams**
- **Annenberg Fellowship with admission to University of Southern California (USC) PhD program, 2019**
- **IEEEExtreme Programming Competition**
  - **IEEEExtreme 10.0, 2016:** Ranked 2<sup>nd</sup> in Turkey, 20<sup>th</sup> in IEEE Region 8, and 65<sup>th</sup> overall
  - **IEEEExtreme 9.0, 2015:** Ranked 3<sup>rd</sup> in Turkey, 30<sup>th</sup> in IEEE Region 8, and 82<sup>nd</sup> overall
- **Nationwide University Entrance Exam (LYS):** Ranked in top 0.01% among 2 million students in Turkey, 2014

## PUBLICATIONS

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**M. T. Avcu, Z. Zhang, and DWS. Chan** “Seizure Detection Using Least EEG Channels by Deep Convolutional Neural Networks” IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019